Memo

To: [Employee Name]
From: Anthony Jordan
Date: 11/4/2025
Subject: Webb Motorsports – UDDS Regenerative Braking and Wheel Speed Sensor Evaluation

---

Dear [Employee],

Webb Motorsports has once more contracted Burdell Inc. to evaluate the wheel-speed sensor package and regenerative braking efficiency of their newest hybrid drivetrain using the Urban Dynamometer Driving Schedule (UDDS) as a test benchmark. The client wants custom numerical codes that can deliver higher accuracy than MATLAB's standard differentiation and integration routines, particularly under realistic sensor noise and dropout conditions. We've been asked to develop, validate, and apply these improved numerical methods to estimate regenerative energy recovery and establish allowable sensor noise limits.

Project Background

Accurate quantification of regenerative braking efficiency depends on two factors: high-quality time–velocity data from reliable wheel-speed sensors and numerical methods capable of differentiating and integrating that data with minimal error, even when spacing is nonuniform or data is noisy. Currently, Webb Motorsports uses MATLAB's built-in forward difference (`diff`) for differentiation and trapezoidal rule (`trapz`) for integration. While simple, these methods accumulate large errors when applied to unevenly spaced or noisy sensor data, particularly in the acceleration and power calculations that underpin regenerative braking estimates.

To improve upon this, the client requests custom-built codes for both operations: a `betterDifferentiator` that maintains $O(h^2)$ accuracy while adapting to unevenly spaced samples, and a `betterIntegrator` that uses Simpson's 1/3 and 3/8 rules when possible but gracefully falls back to trapezoidal rule integration when spacing or dropout prevents their use.

Your analysis will determine: (1) the baseline regenerative braking energy recovered over the UDDS cycle using these improved codes; and (2) the maximum allowable sensor jitter (Gaussian noise) that keeps total distance traveled within 2% error and total regenerative braking energy within 5% error. You will also discuss the impact of dropout and quantization error on these computations—both are critical in evaluating Webb's sensor sourcing options.

Regenerative Braking

The Urban Dynamometer Driving Schedule (UDDS) represents a standardized city-driving profile consisting of 1,369 seconds of speed data including accelerations, stops, and idling events. It is the EPA's regulatory baseline for urban fuel economy and emissions testing and is widely used by vehicle manufacturers to validate hybrid and electric drivetrains.

For this project, you will obtain the dataset directly from the U.S. EPA public FTP repository. UDDS Dataset Link: https://www.epa.gov/vehicle-and-fuel-emissions-testing/dynamometer-drive-schedules. Please note that this data must be converted to [m/s] in the following calculations.

The instantaneous regenerative braking power at any point is:

$$P_{\text{regen}}(t) = \eta \cdot \min(P_{\text{clip}}, \max(0, -m \cdot a(t) \cdot v(t)))$$

It captures how much **mechanical energy**, generated when the vehicle decelerates, is converted back into **electrical energy** and stored in the hybrid drivetrain's battery.

Here's what each term means in context:

- m is the vehicle's mass [kg], used to compute braking force $F = ma(t)$.
- $a(t)$ is the acceleration obtained by differentiating the UDDS velocity data; negative during braking.
- $v(t)$ is the velocity from the UDDS dataset, converted to m/s.
- $-ma(t)v(t)$ is the instantaneous mechanical power associated with braking (force × velocity).
- $\max(0,\cdot)$ ensures only *braking* (negative acceleration) contributes to regenerative energy; acceleration (positive a) yields zero regen power.
- η is the overall regenerative efficiency, accounting for drivetrain and electrical losses.
- P_{clip} is the maximum allowable power [W] imposed by the inverter and battery system. The min() function enforces this physical limit.

Vehicle and Regeneration Parameters	
m	1500 kg
η	0.75
P_{clip}	80 kW

Integrating $P_{\text{regen}}(t)$ over time yields the total recovered energy, $E_{regen}(t)$.

Sensor Noise Background

Wheel-speed sensors, like most real-world measurement systems, are affected by several types of noise that influence the accuracy of derived quantities such as acceleration and regenerative power:

- **Jitter (Gaussian noise, σ):** Random variations in the measured signal amplitude, often caused by electromagnetic interference, vibration, or signal processing limits. In wheel-speed sensors, jitter introduces small random fluctuations in measured velocity that can amplify when numerically differentiated to compute acceleration.
- **Dropout (fraction):** Missing samples in the time series due to transmission errors, timing glitches, or temporary sensor faults. Dropout creates uneven time spacing, forcing numerical algorithms to rely on less accurate trapezoidal or interpolated estimates instead of higher-order Simpson's methods.
- **Quantization error (m/s):** Discretization of continuous speed data into fixed digital levels based on sensor resolution. Coarse quantization can introduce stepwise artifacts in the velocity signal, producing sharp discontinuities and false spikes in derivative estimates.

Together, these effects alter both the smoothness and spacing of data points, affecting the stability and accuracy of numerical differentiation and integration. Your improved algorithms must be designed to mitigate these effects and accurately propagate the physics of braking and regeneration through noisy datasets.

Tasks

- **Task 1 – Develop Improved Numerical Methods:**
  Implement **`betterIntegrator()`** and **`betterDifferentiator()`** functions with adaptive scheme selection and method counters and compare them to basic MATLAB operations that they have been using, namely the **`trapz()`** and **`diff()`** functions for integration and differentiation respectively. Demonstrate the effectiveness of your codes with a validation data set of your choice.
- **Task 2 – Compute Baseline Regenerative Braking Energy:**
  Use the UDDS dataset and the "better" functions to compute baseline braking energy and total distance travelled.
- **Task 3 – Noise Sensitivity and Allowable Jitter:**
  Use `WheelNoiseApp` or your own implementation of the sensor noise equations to simulate sensor noise in measurements conducted under the UDDS, determine the maximum allowable Gaussian noise to meet client specifications, and analyze the effects of dropout and quantization noise sources on the results. See the Appendix provided by our Sensors and Instrumentation Team (SIT) for more details.

Deliverables

- A detailed PowerPoint presentation addressing the tasks above that adheres to client requirements and the Burdell Inc. slide template. **Remember to focus primarily on delivering only data and analyses that are actionable for the client's requests**.

- A .zip file containing your MATLAB scripts and supporting functions that produces all figures and calculations contained in your slides and adhering to the coding guidelines at Burdell Inc. Please make sure that your .zip file contains the following code elements:
    - A script file named **`baselineAnalysis_[last name].m`** that calls all necessary subfunctions and produces all plots used in your presentation document, as well as produces all your numerical results, for the task of establishing baseline measurement values.
    - A script file named **`noiseAnalysis_[last name].m`** that calls all necessary subfunctions and produces all plots used in your presentation document, as well as produces all your numerical results, for the task of using GUI-generated or script generated sensor data to evaluate compliance with the 2% and 5% distance and regenerated energy tolerance thresholds.
    - A function **`[ I,counts ] = betterIntegrator( x,y )`** where **`x`** is the independent variable data, **`y`** is the dependent variable data, **`I`** is the integral value output, and **`counts`** which is a vector of numbers as defined below. The function should examine the step size between data points and determine if there are enough sequential data points of equal spacing to use the Simpson's 1/3 or 3/8 rules. Otherwise, the function will use the trapezoidal rule to perform the integration between steps. The function should keep track of how many applications of each rule are performed and output those numbers into the [3x1] vector of integers **`counts`** which is an output of **`betterIntegrator.m`** and that has the form

      ```
      counts = [
      # of trapezoidal rule applications;
      # of Simpson's 1/3 rule applications;
      # of Simpson's 3/8 rule applications]
      ```

    - A function **`[ D,counts ] = betterDifferentiator( x,y )`** where **`x`** is the independent variable data, **`y`** is the dependent variable data, **`D`** is the vector of derivative values at each value of **`x`**, and **`counts`** is defined below. The function should examine the step size between data points and determine if there are enough sequential data points of equal spacing to use a central divided difference approximation of $O(h^2)$ and use it. Otherwise, the function will use the interpolating polynomials divided difference for unevenly spaced data points of $O(h^2)$ to calculate the approximate derivative. The function should keep track of how many applications of each rule are performed and output those numbers into the vector **`counts`** which is an output of **`betterDifferentiator.m`** and that has the form

```
counts = [
# of central divided difference applications;
# of interp. polynomial divided difference applications]
```

I want you to determine the number of slides and the content of those slides that will make the most compelling case for your conclusions based on the client's needs. Please ensure that your deliverables are clear and concise, as I will be presenting them to the client. Because I haven't performed this analysis myself, I need you to provide detailed presenter notes that explain all your modeling choices, including but not limited to the context under which you are using your numerical frameworks, any relevant information for the client about how they work, and how you analyze the appropriateness and trustworthiness of the results. If you have any questions or need further clarification, do not hesitate to reach out to me.

I look forward to your insightful analysis and contributions to this project.

Best regards,

Anthony Jordan
Engineering Manager
Burdell Inc.